

TradePilot — Bear-Day Audit, Regime Engine & Learnings

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TradePilot — Bear-Day Audit, Regime Engine & Learnings

Session audit log + comprehensive learnings — 2026-06-08 to 2026-06-09

Project	TradePilot (paper-trading engines)
Scope	Trade-audit tooling · v7_regime engine · long/short research · data hygiene
Branch	dev (10 commits)
Trigger	Bear day (2026-06-08): v4 long-only bled, v5 short-tilted profited
Status	Complete — v7 live as paper A/B from 2026-06-09 08:45 IST
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Executive summary

A bear session (Nifty -0.94%) exposed a structural flaw: TradePilot’s scorer emits **zero SELL signals**, so the long-only engine (v4) piled 47 longs into a falling market and bled, while the short-tilted engine (v5) made money. We built a per-trade **counterfactual audit** that quantified the leak (Rs 52,221 left on the table; LONG_IN_BEAR alone -Rs 18,826), ran **adversarially-verified research** into long/short/flip systems, and shipped a **two-layer regime engine (v7_regime)** — a daily ADX/DMI gate that sets the *allowed side*, plus an intraday Supertrend flip within it.

The honest twist: **walk-forward optimization showed the daily gate has no historical edge** (Deflated Sharpe 0.12). Thresholds aren't the lever; the engine's real bet is the *intraday* flip, which has no history to backtest. So v7 went live as a **paper experiment** — the only valid way to test it — graded automatically each day by the audit tool. We also fixed 22 rotted NSE tickers in the universe and added a rot-guard.

Metric	Value	Note
Realized P&L (2026-06-08)	−Rs 11,128	194 trades, 121 long / 73 short
Profit left on the table	Rs 52,221	recoverable with right side + timing
Largest leak (LONG_IN_BEAR)	−Rs 18,826	bigger than the whole net loss
Daily-gate WFO edge	none (DSR 0.12)	thresholds are not the fix
Universe tickers repaired	22 of 29	verified vs live Yahoo before applying
Commits shipped	10	all on dev, tests green

Session audit log (chronological)

#	Phase	What happened	Outcome
1	Launch	Started the market stack for 2026-06-08 open	3 engines + dashboard up
2	Triage	“v4 not scanning” → found it's the intentional 09:30 yfinance warm-up	not a bug
3	Audit build	Built <code>trade-audit.py</code> — 3-layer per-trade counterfactual	root cause found: 0 SELL signals
4	Observe	Started missed-opportunities + profit watchdogs (were never launched)	session recorded live
5	Diagnose	EOD audit: −Rs 11,128, Rs 52k on table, LONG_IN_BEAR dominant	quantified the leak
6	Research	Deep-research harness: 23 sources, 16 claims confirmed / 9 killed	two-layer spec
7	Build	v7_regime engine via TDD + subagent-driven dev	Layer 1 gate + Layer 2 flip, 10 tests
8	Intraday	Wired Layer 2 to live 5-min candles (daily fallback)	true intraday flip, 13 tests
9	Data	Refreshed daily CSVs (452 symbols)	NIFTY-500 current to 2026-06-08

#	Phase	What happened	Outcome
10	Validate	Walk-forward tuning + Deflated Sharpe	honest: no historical edge
11	Go live	v7 = 4th engine, paper A/B, armed for 08:45 auto-launch	first live evidence today
12	Hygiene	Resolved 22 renamed NSE tickers + rot-guard	universe 390/397 fresh

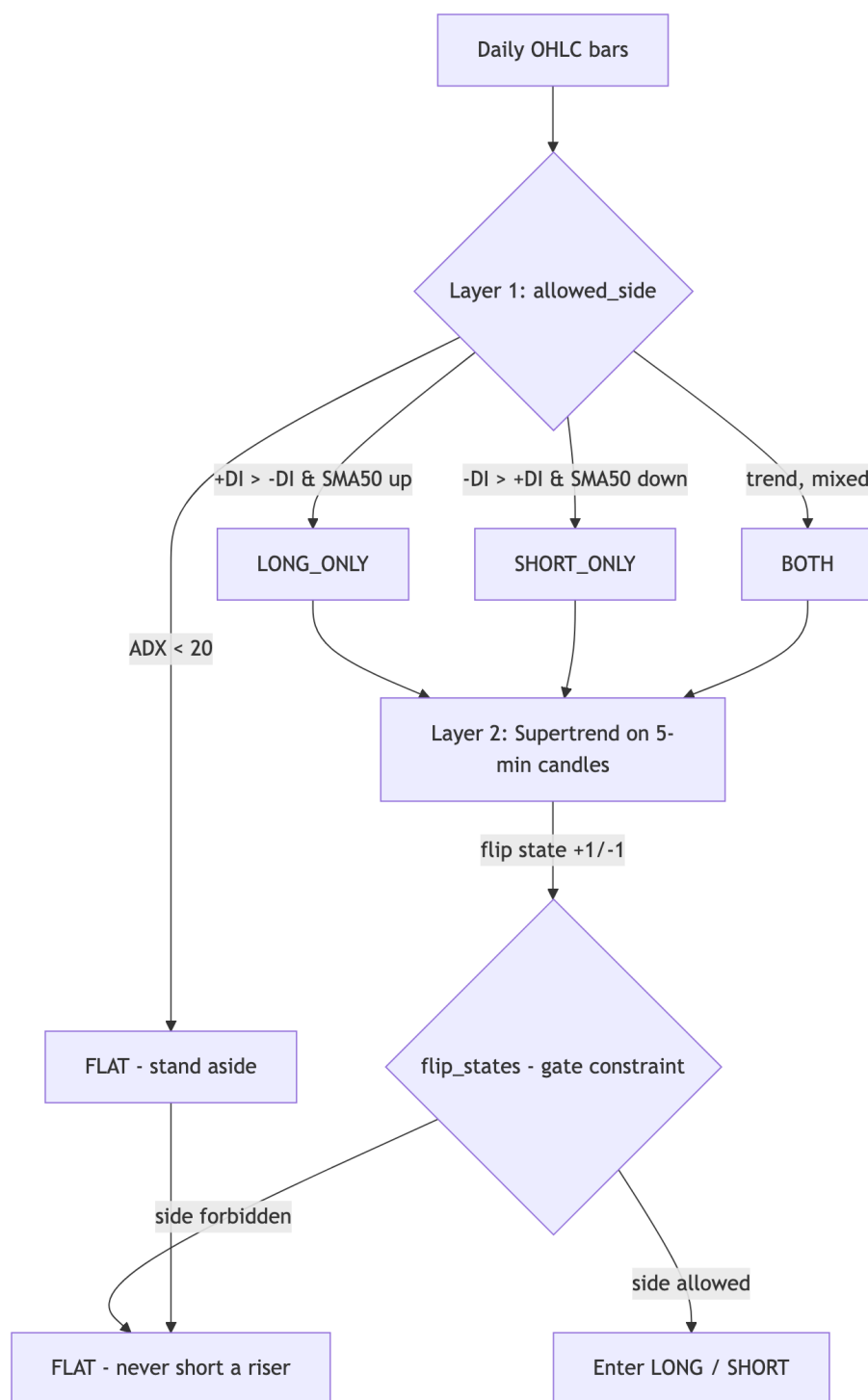
The diagnosis — why the bear day bled

The audit classified every trade. One mistake class explains the entire loss:

Mistake class	Count	Realized	On table
LONG_IN_BEAR	78	−Rs 18,826	Rs 37,650
GOOD_TRADE	57	+Rs 8,245	Rs 6,258
EXIT_TOO_EARLY	35	+Rs 994	Rs 5,205
SHORTED_RISER	22	−Rs 1,535	Rs 3,068
IGNORED_SIGNAL	2	−Rs 7	Rs 40

The natural experiment proved the fix before we built it: **v4 (long-only) −Rs 8,325; v5 (short-tilted, 40 shorts) +Rs 465** on the same falling day. The problem was never stock-picking — it was the absence of a regime gate and a SELL signal.

What we built — the two-layer regime engine



2026-06-09_session-audit-and-learnings-diagram-0

Module	Lines	Purpose
prototype/v7/ regime_gate.py	~70	Layer 1: ADX/+DI/-DI + SMA50 slope → allowed_side
prototype/v7/ supertrend_flip.py	~70	Layer 2: Supertrend SAR + gate-constrained flip machine

Module	Lines	Purpose
scripts/v7_regime-paper-trade.py	(v5 clone)	the engine — direction swapped for gate+flip on 5-min bars
scripts/v7-regime-backtest.py	~60	Layer-1 gated-vs-buy&hold backtest
scripts/v7-wfo-tune.py	~150	walk-forward ADX tuning + Deflated Sharpe
scripts/trade-audit.py	~360	per-trade counterfactual + bear-day solution
scripts/fix-stale-tickers.py	~140	universe rename resolver + rot-guard

Design intent: Layer 1 (slow, daily) decides *which side is even allowed*; Layer 2 (fast, intraday 5-min) decides *when to enter and flip* within it. A side the regime forbids collapses to FLAT — making “short a riser / long a faller” structurally impossible. 13 unit tests pass; the smoke gate stays green.

The honest validation result



2026-06-09_session-audit-and-learnings-diagram-1

Walk-forward optimization across the 49-stock NIFTY-50 basket found **no ADX-threshold combination with a positive out-of-sample Sharpe** (best ≈ -0.10), and a Deflated Sharpe Ratio of **0.12** — essentially no evidence of skill. We did **not** fabricate a tuning win. v7 ships with 25/20 (least-bad) explicitly labelled *experiment, not validated edge*. Because the intraday Layer-2 flip cannot be backtested (no intraday history), the **forward paper A/B is the only honest validation path** — and the audit grades it daily.

Learnings

#	Learning	Why it matters
1	The leak was structural, not stock-picking	No SELL signal + no regime gate → bear days are a long-only bloodbath regardless of which stocks are chosen
2	Counterfactual auditing beats P&L watching	“Rs on the table” + mistake-class breakdown points at the fix; raw P&L only says you lost
3	Single-symbol backtests lie	TCS “+6.6% vs -60%” was regime-cherry-picking; basket WFO + DSR exposed zero edge
4	DSR is the honesty check	A good-looking Sharpe across many tried variants is usually selection luck; DSR 0.12 said “no”
5		

#	Learning	Why it matters
	Some theses can't be backtested	Layer-2 needs intraday history that doesn't exist → forward paper is the only valid test; ship it labelled as an experiment
6	Hardcoded universes rot	29 NSE tickers had renamed/demerged (ZOMATO→ETERNAL, TATAMOTORS demerger); added a --audit rot-guard
7	Verify data mappings before applying	A wrong ticker loads the wrong company — worse than a gap; tested all 22 renames vs live Yahoo first
8	TDD pure modules first	Layers 1 & 2 are pure functions with tests; the engine just wires them — fast, safe, reviewable
9	Right side beats right timing	Flipping every wrong-direction trade would have added +Rs 40,718 vs +Rs 26k from better timing alone
10	Watchdogs must be in the launch path	The observers existed but were never started; capture only began once launched mid-session

Artifacts & commits

Commit	Summary
f4d000b	trade-audit engine — per-trade counterfactual + bear-day solution
295c4bb	two-layer long/short/flip research spec
b64979d	directional indicators (ADX/+DI/-DI)
0e11e45	Supertrend SAR + gate-constrained flip machine
1e92dc7	Layer-1 walk-forward-lite backtest
9f9d7cf	regime-gated paper engine (opt-in)
319472a	Layer 2 flips on intraday 5-min candles
76d3bdd	walk-forward ADX tuner + Deflated Sharpe
18aee44	go live as paper A/B (4th engine)
05e0554	resolve 22 renamed tickers + rot-guard

Reports rendered: docs/audit/2026-06-08_audit-report.pdf, docs/research/2026-06-08_long-short-flip-spec.pdf, docs/superpowers/plans/2026-06-08-v7-regime-engine.pdf, and this document.

Open items / next steps

ID	Item	Priority
NEXT-1	Review v7's first live A/B via tonight's auto-audit (15:35)	high
NEXT-2	If v7 whipsaws, add Supertrend flat-line stand-aside (spec chop guard)	medium
NEXT-3	Add a real SELL tier to the scorer (the actual root-cause fix)	high
NEXT-4	Re-fetch the 7 unresolved tickers periodically (Yahoo coverage gaps)	low
NEXT-5	Wire fix-stale-tickers.py --audit into the weekly cron	low

Bottom line: We turned a losing bear day into a quantified diagnosis, a researched fix, a shipped experiment, and a cleaner data foundation — while staying honest that the edge is unproven and today is the first real test.